ROYAL GLOBAL UNIVERSITY
GUWAHATI

(ROYAL SCHOOL OF MEDICAL & ALLIED SCIENCES)
(RSMAS)

DEPARTMENT OF ANESTHESIA
&
OPERATION THEATRE TECHNOLOGY

Course structure & Syllabus
In Line with National Commission for Allied and Healthcare Professions
(NCAHP)
For
MASTERS OF ANESTHESIA & OPERATION THEATRE TECHNOLOGY

W.E.F 2026 – 27

Curriculum Masters of Anesthesia & Operation Theatre Technology (2 years program)

PREAMBLE

The **Department of Anesthesia and Operation Theatre Technology**, Royal School of Medical and Allied Sciences (RSMAS), The Assam Royal Global University, is dedicated to advancing excellence in postgraduate education, clinical practice, research, and innovation in perioperative and anesthesia technology. Guided by the Vision and Mission of The Assam Royal Global University and aligned with the provisions of the **National Commission for Allied and Healthcare Professions (NCAHP) Act, 2021**, the Masters of Anesthesia and Operation Theatre Technology is designed to develop highly competent professionals capable of delivering advanced, safe, ethical, and evidence-based anesthesia and operation theatre services across diverse healthcare settings.

The curriculum adopts the principles of **Competency-Based Education (CBE)** and **Outcome-Based Education (OBE)**, integrating advanced biomedical sciences, anesthesia technology, perioperative patient management, critical care concepts, surgical assistance, research methodology, healthcare quality assurance, and interdisciplinary collaboration. Emphasis is placed on developing higher-order clinical reasoning, decision-making, patient safety, infection prevention and control, healthcare ethics, leadership, communication, digital healthcare competencies, and lifelong learning to meet the evolving demands of modern surgical and perioperative care.

The programme promotes experiential learning through structured clinical postings, simulation-based education, case-based discussions, seminars, research projects, and evidence-based clinical practice. Graduates will be equipped to function effectively as advanced anesthesia and operation theatre technologists, educators, researchers, clinical supervisors, and healthcare leaders, contributing significantly to quality improvement, patient safety, surgical outcomes, and the advancement of allied healthcare services.

The Department envisions becoming a centre of excellence in postgraduate anesthesia and operation theatre education by nurturing professionals distinguished by clinical expertise, ethical integrity, scientific inquiry, innovation, and compassionate patient care. Through academic excellence, research, and professional collaboration, the programme seeks to prepare graduates capable of addressing contemporary healthcare challenges while contributing to national healthcare priorities and global standards of perioperative practice.

"Committed to Excellence in Advanced Anesthesia and Perioperative Care through Education, Research, Innovation, Leadership, and Compassionate Service to Society."

This syllabus was reviewed and approved by the **11th School Board of Studies (BoS)** of the Royal School of Medical and Allied Sciences (RSMAS) and subsequently ratified by the **___ Academic Council** of The Assam Royal Global University, Assam, and shall be implemented with effect from the **Academic Session 2026-27**.

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1.0. Masters of Anesthesia & Operation Theatre Technology: Curriculum Background

The need for accuracy in preparation and delivery is a critical component of modern technology driven healthcare and requires knowledge and understanding of the basic sciences as well as the interaction between the technology used in operation theatres and the site within the body that needs the surgical intervention. In an era of greater complexity of technology and techniques, the role of the Anaesthesia & operation theatre technologist (AOTT) and his/her level of responsibility is continually evolving and expanding. Education programmes should provide the AOTT with the scientific theoretical foundation of the profession and enable them, as practitioners, to be able to synthesize, evaluate and apply their knowledge in a clinical setting.

The aims of the recommended curriculum are to produce graduates who are:

i. Technically and clinically competent;
ii. Aware of safety issues and the importance of quality assurance;
iii. Understand the theoretical basis for evidence based practice;
iv. Effective members of the multidisciplinary team;
v. Prepared to participate in or initiate research into practice;
vi. Can work according to registration requirements on the respective continents.

All aspects of anaesthesia & operation theatre technology have been considered in the development of this curriculum together with the identification of the roles expected for different levels based on their qualification and experience. The need for connecting the dots between the education and employment practices has been the road map for devising this curriculum.

Foundation course has also been designed to bring all the students at the same level of understanding with respect to basic healthcare related norms before the start of a career in a healthcare professional course. The foundation course is mandatory for all the allied and healthcare professional courses and for both entry level courses – diploma as well as degree. If a diploma holder has completed the foundation course and is willing to pursue the degree course, the candidate will directly get entry for next semester, however a pre- qualifier skill test will have to be satisfactorily completed, if not, then the candidate will have to undergo the first semester of foundation course again.

1.1 Introduction:

An Anaesthesia and Operation Theatre (A&OT) Technologist forms an intrinsic part of any hospital. To become a trained professional one must undertake Anaesthesia and Operation Theatre Technology (A&OTT) program. This program aims at providing advanced knowledge in the field of Anaesthesia & Operation Theatre Technology. It aims to provide graduates who are well versed in planning, setting up, maintaining and managing the anaesthesia and surgical procedures. This program provides training on working and handling of anaesthesia and surgical equipment in and outside the Operating Room(OR). This program will bring out post graduates who are qualified and eligible to work as clinical supervisors in the Operation Theatre (OT), teach and train the students in the same specialty.

Learning Objectives: At the completion of this course, the student should be –

i. Able to help the anesthesiologist in administering anesthesia, assist in various procedures and also help in continuous monitoring of patients during surgery.
ii. Able to train and develop an individual to independently handle the latest technology and high end biomedical equipment in Operation Theatre
iii. Able to assist anesthesiologists in developing patient care plans, including pre-operative, surgical theater, recovery room, and post-operative intensive care procedures.

iv. Able to do- patient data collection, catheter insertion, airway management, assisting the administration and monitoring of regional and peripheral nerve blockades, support therapy, adjusting anesthetic levels during surgery, inter-operative monitoring, postoperative procedures, pain clinics and patient education, and administrative tasks.
v. Able to assist surgeons during perioperative period
vi. Able to manage medical gases and pipeline system
vii. Able to assist in Intensive care unit
viii. Able to manage Central sterile supply department
ix. Able to apply principles of management in health care setting
x. Able to apply statistical methods for research and analysis

1.2 Scope of Practice:

i. The Anaesthesia & Operation Theatre Technologists mainly assists the Anaesthesiologist and Surgeons in preparing and managing the patient for surgery during pre-operative, intra-operative and post-operative periods
ii. Responsible for maintaining anaesthesia records
iii. Understand the anxiety of patient in preoperative state and appropriately assist to shift, induce anaesthesia and position patient for surgical procedure
iv. Assist anesthesiologists and surgeons with Non-Operating Room Anaesthesia(NORA) and surgery, like in CT and MRI suits, dental clinic, Labor theatre, IVF centres, burn centres, endoscopy room, and Cardiac catheterization laboratory, pain relief procedures etc.
v. Assist anesthesiologists during management of patient in the emergency departments and intensive care units
vi. Assist consultant in transportation of the patients
vii. Responsible for managing an OR as clinical supervisor
viii. Postgraduates are qualified to teach at the university level and responsible for developing curriculum and delivering the course content, grading assignments, assessing the students and providing feedback and support the students
ix. Postgraduates can also be involved in research related to their field of study, in addition to teaching and other academic responsibilities
x. There is a scope to get a fellowship or Ph. D after post-graduation
xi. Assist disaster team in disaster situations and national emergencies on field and safe transport in ambulance.

1.3 Job Responsibilities for MAOTT:

- **As a Clinical Supervisor/ OT Manager:**

a. Providing supervision to clinical staff in the OT
b. Monitoring progress and ensuring that clinical staff adhering to ethical and legal standards
c. Maintaining the records, inventory and human resource in the OT

- **As an Anaesthesia/Surgical assistant:**

a. Assisting Anaesthesiologist and surgeons during surgeries and anaesthetic procedures for general surgeries and super specialties like Cardio Thoracic surgeries, Neuro surgeries, Pediatric surgeries, Renal surgeries, Plastic and reconstructive surgeries, Onco-surgeries, Ophthalmic surgeries, Otorhinolaryngology (ENT) surgeries, Orthopedic surgeries and OBG procedures
b. Preparation and maintenance of advanced anesthetic equipment including fiberoptic

bronchoscopes, ultrasound used in anaesthesia, advanced airway gadgets, invasive monitoring devices and infusion pumps

c. Maintaining the stock and records of anaesthetic drugs including narcotics
d. Preparation and maintenance of advanced surgical equipment including laparoscopes, endoscopes, microscopes, electrosurgical units, laser and robotic surgical instruments

- **As Teaching faculty**

a. Designing course curriculum, developing and implementing new teaching strategies and methods
b. Student assessment
c. Mentoring the students
d. Organizing and participating in departmental meetings, workshops, conferences, faculty development programmes and other academic activities
e. Conducting research related to their field of study
f. Collaborating with researchers and industry experts.
g. Serving on academic committees and participating in other service activities

1.4 Eligibility for admission:

A candidate seeking admission to Master"s in Anaesthesia & Operation Theatre Technology program must have passed Bachelor"s degree of minimum of 3 years duration in Anaesthesia & Operation Theatre Technology/Anaesthesia Technology/ Operation Theatre Technology from a recognized university, with minimum 50% aggregate marks at HSC for open category and minimum 45% aggregate marks for reserve category.

1.5 Duration of the course:

The duration of Master"s in Anaesthesia & Operation Theatre Technology shall be two academic years (4 semesters)

1.6 Medium of instruction:

English shall be the medium of instruction for all the subjects of study and for examination of the course.

1.7 Attendance:

- 80% attendance separately in theory and practical/Clinical postings, in each course.

1.8 Assessment and Evaluation:

- **Scheme of Evaluation**
  - The academic performance is assessed on the basis of both Continuous Internal Evaluation (CIE) assessment and End Semester Examination (ESE) in each semester.
  - ESE weightage will be in the ratio of 30 % for CIE and 70 % for ESE.
- **Continuous Internal Evaluation (CIE)**

The internal assessment for theory and practical's shall be 20 marks each.

Internal assessment for theory shall be calculated as average of two sessional examinations, seminars, assignments etc. The internal assessment for practicals shall be calculated on the basis of the records of the practicals maintained and examinations.

A Candidate must secure at least 40% of total marks fixed for internal assessment in a particular course in order to be eligible to appear in the university examination of that course. If candidate is

having inadequate internal assessment marks shall appear for internal assessment in the next semester.

- **End semester examination (ESE)**

  A candidate who satisfies the requirements of attendance, progress and conduct shall be eligible to appear for the university examinations. There shall be a university examination at the end of each semester.

  To be eligible to appear for University examination a candidate should fulfil all the following conditions:

  i. 80% attendance separately in theory and practical/Clinical postings, in each course.
  ii. Secure at least 40% of total marks fixed for internal assessment in a particular course in theory and practical separately.
  iii. Submit dissertation in the prescribed format, as per the stipulated schedule in IV semester.

- **Dissertation:**

  A candidate is required to carry out research study in a select area of his subject, under the supervision of a faculty guide. The results of such a study shall be submitted to the University in the form a dissertation as per the prescribed format and within the date stipulated by the University.

- **Guide:**

  A guide shall be MD/DNB/PG Diploma/M. Sc. or equivalent qualification as recognized by the University in relevant subject with two years‟ experience in respective Allied Health Science Course. In case of postgraduate diploma, he/she should have two publications in indexed journals. However, a Co-guide can be opted wherever required with prior permission of the Institute and University. The Co-guide shall also be a postgraduate teacher recognized by the University as a guide.

  Candidate shall submit synopsis to the University through the Guide and Head of the Institute, on or before the completion of one month of second semester, or within the date notified by the University, whichever is earlier.

  Once the synopsis is approved and registered by the University, no change in the topic or Guide shall be made without the prior approval of the University.

  In the event of registered Guide leaving the Institute or in the event of the death of the guide, a change of Guide shall be permitted by the University, on the specific recommendation of the Institute.

- **Ethical clearance**

  Ethical clearance should be obtained for a study involving any procedure on human subjects. The candidate should apply for the certificate to the Ethics Committee of the respective Institute/University, through the Guide and present the study before the Committee for clearance. A copy of the certificate should be attached along with the synopsis forwarded at the time of submission of synopsis. All such clearance should be sought within one month of commencement of II semester.

- **Submission of synopsis**

  Synopsis should be verified by guide, HOD and departmental curriculum development cell and approved by the institutional ethics committee before submission to the university. The synopsis should be submitted as per the format, on or before one month of second semester, or within the date notified by the University, whichever is earlier. Once the synopsis is approved and registered by the university no change in the topic or Guide shall be made without the prior approval of the University.

- **Final submission of the dissertation**
  The dissertation complete in all respects and duly certified by the Guide/Co-guide, Course Co-ordinator/ HOD/ Director should be submitted it to the Controller of Examinations as per the date specified by the respective University, generally three months before commencement of University examination.

1.9 Semester & Credit System Regulations:

Definitions of Key Words:

Credit: A unit by which the course work is interpreted. It functions the number of hours of instructions required per week.

- **Credit Distribution**

Each semester would get a complete 15 weeks for academics, excluding sessional exams, study leave, university exams, semester break, declared holidays and non-academic events.

- Lectures (L)/Theory (T): 1 hour /week = 1 credit
- Practical (P)/ Demonstration (D): 2 hours/week = 1 credit
- Clinical Posting (CLP): 3 hours/week = 1 credit

- **Credit distribution table format:**

(L - Lectures, T - Theory, P/D –Practicals/Demonstration, CLP – Clinical Posting, CR - Credits. IA - Internal Assessment, SEE – Semester End Examination)

- **Semester Grade Point Average (SGPA):** The overall performance of a student in each semester
- **Grade Point:** It is a numerical marking allotted to each letter grade on a 10-point scale.

- **Grading and Classification:**

Letter grades and CGPA

The performance of a candidate shall be evaluated according to a letter grading system, based on both CIE and ESE. The letter grades (O, A+, A, B+, B, C, F and AB) indicate the level of academic achievement assessed on a 10-point scale (0 to 10) (see table below).

- A candidate shall be considered to have completed a course successfully and earned the credits assigned if he secures an acceptable letter grade in the range O-C.
- Letter grade 'F' in any course implies failure in that course and no credit is earned in such a case.
- A candidate having satisfactory attendance at classes and meeting the passing standard at CIE in a course, but remaining absent from, ESE shall be awarded 'AB' grade in that course.

Marks Range (%)	Grade Point	Letter Grade	Descriptor
90 & above	10	O	Outstanding
80 -89	9	A+	Excellent
70-79	8	A	Very Good
60-69	7	B+	Good
55-59	6	B	Above Average
50-54	5	C	Average
Below 50	0	F	Fails
Absent	0	AB	Absent

Classification	CGPA
First Class with Distinction	7.00 and above
First Class	6.00-6.99
Second Class	5.50-5.99
	5.00-5.49
Fail	Less than 5.0

- **Grade Point Averages:**
  The overall performance of a candidate will be indicated by Cumulative Grade Point Average (CGPA). For each course grade points will be awarded as per a letter grading system. As shown in table above.

- **Semester Grade Point Average (SGPA)** is computed as follows:

  ∑ [(course credit) X (Grade point)] for all courses with Letter grades, including F
  SGPA= ----------
  ∑ [(course credits)] for all courses with Letter grades, including F

- **Cumulative Grade Point Average (CGPA)** is computed as follows:

  ∑ [(course credit) X (Grade point)] for all courses for all semesters with, Letter grades excluding F
  CGPA= ----------
  ∑ [(course credits)] for all courses for all semesters with Letter grades, excluding F

- **Conversion of Grades into Percentage:**
  Formula for conversion of CGPA into percentage:
  CGPA earned x 10 = Percentage of marks scored Illustration: (CGPA Earned 8.18 x 10) = 81.80 %

  Award of Class:

  The candidate, who has passed all the courses prescribed, shall be declared to have passed the program. Class will be awarded only to those who pass the entire examination in the first attempt and on the basis of the aggregate of marks scored in individual semester.

A candidate shall be considered to have completed a course successfully and earned the credits assigned, if he secures an acceptable letter grade in the range O-C. Letter grade „F" in any course implies failure in that course and no credit is earned.

A candidate having satisfactory attendance at classes and meeting the passing standard at CIE in a course, but remained absent from SEE shall be awarded „I" grade in that course.

- A candidate who secures GPA ≥ 7.00 and above in first attempt shall be declared to have passed in „First Class with Distinction".
- A candidate who secures GPA ≥ 6.00 or more but less than 7.00 in the first attempt shall be declared to have passed in „First Class".
- A candidate who secures GPA ≥ 5.00 or more but less than 6.00 in the first attempt shall be declared to have passed in „Second Class".
- A candidate who secures GPA ≥ 4.00 or more but less than 5.00 in the first attempt shall be declared to have passed in „Pass Class".
- Candidates who pass the examinations in more than one attempt shall be declared as passed in „Pass" class irrespective of the percentage of marks secured.
- An attempt means the appearance of a candidate for one or more courses either in part or full in a particular examination. If a candidate submits application for appearing for the examination but does not appear for any of the courses either in full or part in the university examination, he can appear for supplementary examination provided other conditions such as attendance requirement, internal assessment marks, etc are fulfilled and his appearing in the supplementary examination shall be considered as the first attempt.
- Candidates who pass the subjects in the supplementary examinations are not eligible for the award of Gold Medal or Merit Certificate.

Carry Over Benefit:

A candidate shall appear for all the subjects of that particular semester in the University examinations but failed in that semester can avail this benefit provided:

- A candidate who fails in I semester is allowed to move to II semester. The candidates with back log subjects shall take both I semester backlog papers as well as II semester papers. Candidate with a backlog of not more than 2 papers in I & II semester put together is allowed to go to the III semester.
- Candidates who have failed in not more than 2 subjects of II semester and III semester (put together) and not having backlog of I semester papers are only permitted to go to IV semester.
- The candidate is permitted to appear for the IV semester examination along with the backlog subjects of II and III semesters and should pass all the subjects, including the backlog subjects to be declared as having completed the course.
- However, a candidate should pass all the lower semester papers before the announcement of final semester results.

Supplementary Examinations:

Supplementary examination shall be conducted by the university for the benefit of unsuccessful candidates. Lower semester examinations shall be conducted by the University along with current semester examinations for the benefit of unsuccessful candidates.

- A Candidate detained for lack of attendance will be barred from appearing in any one or all course/s for the supplementary examination.
- A candidate permitted to appear for the supplementary examination can improve his internal assessment marks before he takes the supplementary examination by subjecting himself to internal assessment.

1.10 SCHEME OF STUDY : BACHELOR OF ANESTHESIA & OPERATION THEATRE TECHNOLOGY:

First Semester:

FIRST SEMESTER												
Course Code	Course Title	Credit(s) distribution				Marks Distribution						
		L/T	P/D	CLP	CR	Theory			Practical			Total
						IA	SEE	Total	IA	SEE	Total	
MAOT1.1	Review of Applied sciences	3	--	--	3	20	80	100	---	--	--	100
MAOT1.2	Anaesthetic equipment and procedures	3	1	2	6	20	80	100	20	80	100	200
MAOT1.3	Anaesthetics and emergency drugs	3	1	2	6	20	80	100	20	80	100	200
MAOT1.4	Applied surgical technology	3	--	2	5	20	80	100	--	--	--	100
GRAND TOTAL					**20**							**600**

L - Lectures, T - Theory, P/D –Practicals/Demonstration, CLP – Clinical Posting, CR - Credits. IA - Internal Assessment, SEE – Semester End Examination

Second Semester:

SECOND SEMESTER												
Course Code	**Course Title**	**Credit(s) distribution**				**Marks Distribution**						
		L/T	**P/D**	**CLP**	**CR**	**Theory**			**Practical**			**Total**
						IA	**SEE**	**Total**	**IA**	**SEE**	**Total**	
MAOT 2.1	Advanced anaesthesia techniques	3	1	2	6	20	80	100	20	80	100	200
MAOT 2.2	Speciality Anaesthesia-I	3	1	1	5	20	80	100	20	80	100	200
MAOT 2.3	Clinical surgery-I	3	1	2	6	20	80	100	20	80	100	200
MAOT 2.4	Research methodology & biostatistics (should include one unit related to teaching methodology)	4	--	--	4	20	80	100	---	---	--	100
GRAND TOTAL					**21**							**700**
L - Lectures, T - Theory, P/D –Practicals/Demonstration, CLP – Clinical Posting, CR - Credits. IA - Internal Assessment, SEE – Semester End Examination												

Third Semester:

SECOND SEMESTER												
Course Code	Course Title	Credit(s) distribution				Marks Distribution						
		L/T	P/D	CLP	CR	Theory			Practical			Total
						IA	SEE	Total	IA	SEE	Total	
MAOT 2.1	Advanced anaesthesia techniques	3	1	2	6	20	80	100	20	80	100	200
MAOT 2.2	Speciality Anaesthesia-I	3	1	1	5	20	80	100	20	80	100	200
MAOT 2.3	Clinical surgery-I	3	1	2	6	20	80	100	20	80	100	200
MAOT 2.4	Research methodology & biostatistics (should include one unit related to teaching methodology)	4	--	--	4	20	80	100	---	---	--	100
GRAND TOTAL					21							700
L - Lectures, T - Theory, P/D –Practicals/Demonstration, CLP – Clinical Posting, CR - Credits. IA - Internal Assessment, SEE – Semester End Examination												

Fourth Semester:

FOURTH SEMESTER												
Course Code	Course Title	Credit(s) distribution				Marks Distribution						
		L/T	P/D	CLP	CR	Theory			Practical			Total
						IA	SEE	Total	IA	SEE	Total	
MAOT4.1	Clinical Medicine & critical care management	5	1	1	7	20	80	100	20	80	100	200
MAOT4.2	Dissertation evaluation	--	--	8	8	--	---	--	-	100	100	100
GRAND TOTAL					15							300
L - Lectures, T - Theory, P/D –Practicals/Demonstration, CLP – Clinical Posting, CR - Credits. IA - Internal Assessment, SEE – Semester End Examination												

MODEL CURRICULUM OUTLINE

First Semester

FIRST SEMESTER					
Course code	**Course Title**	**Total hours of instruction**			
		Theory	**Practical**		**Total**
			Demonstration	**Clinical Posting**	
MAOT1.1	Review of applied sciences	45	--	--	**45**
MAOT1.2	Anaesthetic equipment and procedures	45	30	90	**165**
MAOT1.3	Anaesthetics and emergency drugs	45	30	90	**165**
MAOT1.4	Applied surgical technology	45	--	90	**135**
TOTAL		**180**	**60**	**270**	**510**

Second Semester

SECOND SEMESTER					
Course code	**Course Title**	**Total hours of instruction**			
		Theory	**Practical**		**Total**
			Demonstration	**Clinical Posting**	
MAOT2.1	Advanced anaesthesia techniques	45	30	90	**165**
MAOT2.2	Speciality Anaesthesia-I	45	30	45	**120**
MAOT2.3	Clinical surgery-I	45	30	90	**165**
MAOT2.4	Research methodology & biostatistics (should include one unit related to teaching methodology)	60	--	--	**60**
TOTAL		**195**	**90**	**225**	**510**

Third Semester

THIRD SEMESTER					
Course code	**Course Title**	**Total hours of instruction**			
		Theory	**Practical**		**Total**
			Demonstration	**Clinical Posting**	
MAOT3.1	Speciality Anaesthesia-II	45	30	180	**255**
MAOT3.2	Clinical surgery-II	45	30	135	**210**
MAOT3.3	Health care management(includes principles of management)	45	--	--	**45**
TOTAL		**135**	**60**	**315**	**510**

Fourth Semester

FOURTH SEMESTER					
Course code	**Course Title**	**Total hours of instruction**			
		Theory	**Practical**		**Total**
			Demonstration	**Clinical Posting**	
MAOT4.1	Clinical Medicine & critical care management	75	30	45	**150**
MAOT4.2	Dissertation- Project work	--	--	360	**360**

1.11. MAOTT Curriculum:

First Semester

REVIEW OF APPLIED SCIENCES
Theory: 45 hours
OBJECTIVES:
At the end of this semester students should be able to:

1. Describe the parts of brain
2. Mention the coverings of spinal cord & CSF
3. Illustrate the anatomical structure of the bronchi and their subdivisions.

4. Explain how the blood transports oxygen and carbon dioxide.
5. Describe gas exchange in the pulmonary and systemic circuits.
6. Understand the clinical physiology of ANS and α and β receptors
    a. Uses and side effects of major drugs in each category.
    b. Clinical circumstances where these agents may be beneficial.
7. Ability to identify, address and treat the associated risk factors according to evidence based guidelines.
8. Understand the pharmacology of diuretics, antihypertensives, antiarrhythmics, corticosteroids, drugs used for coagulation disorders and other hormonal preparations
9. Understand hypersensitivity and antibiotic sensitivity
10. Understand and manage nosocomial infections

COURSE OUTCOMES:

At the end of the course, students will be able to...

- CO1: Discuss the anatomy of Brain & spinal cord
- CO2: Discuss the anatomical & physiological aspects of respiratory and cardiovascular system.
- CO3: Record and interpret the lung function tests.
- CO4: Explain the homeostasis and hemostasis
- CO5: Describe the pharmacology of Adrenergic and Cholinergic drugs and their blockers.
- CO6: Explain the pharmacology of Diuretics, antiarrhythmics,antihypertensives and corticosteroids.
- CO7: Describe the pharmacology of drugs used in coagulation disorders and other drugs
- CO8: Understand immunology & nosocomial infections.

Unit-I: Applied Anatomy
Brain & Spinal Cord

- Parts of brain
- Covering of spinal cord
- Tracts of spinal cord
- CSF

B. Broncho pulmonary segments and alveoli
C. Systemic and pulmonary circulation
D. Coronary circulation

E. Hepatobiliary system
    - Liver
    - Biliary apparatus

F. Urinary system

Unit-II: Applied Physiology

A. Respiratory physiology
    - Spirometry
    - Lung function tests
    - Oxygen transport & CO2 transport

B. Conducting system of the heart

C. Neuro muscular junction

D. Coagulation and its disorders

E. Acid base physiology

F. Hepatobiliary physiology

Unit-III: Applied Pharmacology

A. Adverse drug reactions & Drug interactions

B. Drugs acting on autonomic nervous system
    - Adrenergic drugs & blockers
    - Cholinergic drugs & blockers

C. Diuretics

D. Antiarrhythmics

E. Antihypertensive drugs

F. Corticosteroids

G. Drugs used in disorders of coagulation

H. Others
    - Insulin
    - Oxytocin
    - Methylergometrine
    - Prostodine
    - Antacids

Unit-IV: Microbiology

A. Immunology
    - Immunity
    - Antigen
    - Antibody
    - Hypersentivity
    - Antibiotic resistance

B. Nosocomial infection – Introduction, prophylactic immunization & management
    - Meningitis
    - Hepatitis

- HIV infection & AIDS
- UTI
- Pneumonia
- MRSA infection

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Review of applied sciences** shall be as given under.

Type of questions	**No. of questions**	**Marks**	**Total**
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20 marks

There shall be no practical examination

Reference books:

1. Essentials of medical pharmacology (Latest edition) – KD Tripathi
2. Textbook of Pharmacology for Dental and Allied Health Sciences (Latest edition)- Padmaja Udaykumar
3. Text book of Medical Physiology – Indu Khurana
4. Manipal manual of Anatomy for Allied Health Sciences Courses- Sampath Madhyastha
5. BD Chaurasia‟s Human Anatomy – volume1
6. Text book of microbiology- C P Baveja

FIRST SEMESTER

ANAESTHETIC EQUIPMENT AND PROCEDURES

Theory: 45 hours

Practical/Demonstration: 30 hours

OBJECTIVES:

At the end of the first semester students should be able to

1. Check adequacy of gas supplies, indent for required supplies, and ensure safe storage and commissioning of gas supplies.
2. Perform routine and periodic checks of
    a. Anaesthesia workstation
    b. Gases supply system
    c. Scavenging system
3. Trouble shoot problems in gas supply, anaesthesia workstation and scavenging system.
4. Ensure availability of standard monitors and their proper functioning.
5. Able to trouble shoot minor complications. Periodic check and suitability of monitoring. Competency in basic interpretation of monitored values.
6. Understand simulation in anaesthesia

COURSE OUTCOMES

At the end of the course, students will be able to…
CO1: Understand & demonstrate anaesthesia workstation, its components & safety CO2:
Practice safety check of anaesthesia workstation.
CO3: Discuss and operate monitoring devices
CO4: Understand and operate airway devices
CO5: Prepare difficult airway cart.
CO6: Set up for hemodynamic monitoring.
CO7: Understand types of simulators and simulation in anaesthesia

Unit-1

A. Medical Gases and Distribution System
    - Medical gas supply, storage and safety

B. The modern integrated Anaesthesia workstation
    - Anaesthesia machine & its components
    - Fail safe system
    - Safety check of anaesthesia machine
    - Scavenger system

Unit-2

C. Monitoring Equipment

- Respiratory gas monitoring and minimum alveolar concentration
- Equipments to measure depth of anaesthesia
  - Bi-spectral index
  - Entropy
- Neuromuscular block monitoring equipment
- Cardiac output monitors
- Equipment for central neuraxial and regional blocks
  - Needles
  - Catheters
  - Nerve locators
  - Ultrasound device
- Anesthesia equipment for magnetic resonance imaging
- Equipment for anaesthesia in remote locations

Unit-3

D. Airway gadgets and their accessories

- Surgical airway equipments
- Percutaneous airway equipments
- Optical laryngoscopes
- Fiberoptic bronchoscopes
- Airway introducers
- Alternative to intubation
- Gadgets for difficult airway

Unit-4

E. Hemodynamic monitoring

- Pressure transducers: resonance
- Damping
- Invasive & non-invasive blood pressure measurement
- Oscillometer

F. Defibrillator

G. Updates in anaesthesia devices

- Advanced Supraglottic Airway Devices
- Ventrain device
- Optical Stylets
- Smart pumps

H. Simulation in anaesthesia

Practical/Demonstration:

- Perform routine and periodic checks and troubleshooting of anaesthesia workstation, Gas Supply and Scavenging system
- Analyze the proper working of monitoring systems
- Assembly of gadgets for difficult airway management
- Preparation of invasive blood pressure monitoring systems
- Arrangement of trolley for surgical airway
- Simulation in anaesthesia – Mask ventilation, laryngoscopy and endotracheal intubation
- Training of use of anaesthesia equipment in MRI field
- Pre-use check of anaesthesia equipment
- Sterilization and maintenance of anaesthesia equipment
- Anaesthesia in remote locations- Exposure to patients in
    - Emergency department
    - Endoscopy room,
    - IVF center,
    - Labour theatre

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Anaesthetic Equipment & Procedures** shall be as given under.

Type of questions	**No. of questions**	**Marks**	**Total**
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory internal assessment-20marks

Practical examination-100marks

Marks pattern for practical examination:-

Practical exercises		**Viva- Voce**	**Internal assessment**
Skill	**Response**		
30	30	20	20
Grand Total- 100			

Reference books:

1. A practical approach to anaesthesia equipment- Jerry A Dorsch & Susan E Dorsch
2. Anaesthesia equipment simplified- Gregory Rose & J Thomas Mclarney
3. Understanding anaesthetic equipments and procedures A practical approach- Dwarakadas K Baheti & Vandana V Laheri

FIRST SEMESTER

ANAESTHETICS AND EMERGENCY DRUGS

Theory: 45 hours

Practical/Demonstration: 30 hours

OBJECTIVES:

At the end of the semester students should be aware

1. Preparation, strength, dose, dilution and storage of commonly used drugs during anaesthesia and resuscitation.
2. Common indications and main pharmacological effects of drugs, fluid and blood used in anaesthesia.
3. Common side effects of the drugs.
4. Identification and immediate treatment of common side effects.

COURSE OUTCOMES

At the end of the course, students will be able to...

- CO1: Describe the pharmacology of emergency drugs
- CO2: Group and demonstrate preparation of emergency drugs. CO3: Discuss and use antiemetic drugs
- CO4: Demonstrate and dilute electrolytes
- CO5: Demonstrate fluid infusion including blood
- CO6: Discuss transfusion hazards

Unit-1

- Core drugs in anaesthetic practice
  - General anaesthetic agents
    - IV inducing agents
    - Opioid analgesics & antagonists
    - Volatile anaesthetic agents
    - Sedative hypnotics
  - Local anaesthetics
  - Muscle relaxants and anticholinesterases
  - NSAIDS

Unit-2

- Emergency drugs
  - Oxygen
  - Inotropes
  - Adrenaline
  - Noradrenaline
  - Dopamine
  - Dobutamine
  - Isoprenaline
  - Vasopressine
  - Digoxin

- Sympathomimetics
  - Mephentermine
  - Phenylephrine
  - Ephedrine

- Vasodilators
  - Sodium Nitroprusside
  - Nitroglycerin
  - Nifedipine

- Anticholinergics
  - Atropine
  - Glycopyrrolate

- Bronchodilators
  - Deriphylline
  - Aminophylline
  - Salbutamol

Unit-3

- Other important drugs
  - Antiemetics and related drugs
  - Antimicrobials
  - Antihistamines
  - Electrolytes
    a. Sodium Bicarbonate
    b. Calcium Gluconate
    c. Calcium Chloride
    d. Magnesium Sulphate
    e. Potassium Chloride

Unit-4

- Fluid therapy and transfusion
  - Distribution of body fluids
  - Crystalloids
  - Colloids
  - Fluid infusion and effect on body fluid compartments
  - Blood components
  - Blood transfusion
  - Transfusion hazards

Practical/Demonstration:

- Preparation, strength, dosage, dilution and labelling of
  - General anaesthetic agents
    - IV inducing agents

- Opioid analgesics & antagonists
- Volatile anaesthetic agents
- Sedative hypnotics

- Local anaesthetics
- Muscle relaxants and anticholinesterases
- NSAIDS
- Oxygen
- Inotropes
- Sympathomimetics
- Vasodilators
- Anticholinergics
- Bronchodilators
- Antiemetics
- Electrolytes

- I. V fluid infusion for adults and paediatric patients
- Blood transfusion demonstration
- Identification and immediate treatment of common side effects of drugs

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Anaesthetics and Emergency drugs** shall be as given under.

Type of questions	**No. of questions**	**Marks**	**Total**
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20 marks

Practical examination- 100marks
Marks pattern for practical examination: -

Practical exercises		**Viva- Voce**	**Internal assessment**
Skill	**Response**		
30	30	20	20
Grand Total- 100			

Reference books:

1. Comparative Pharmacology for Anaesthetist: Armeen Ahmed, Vipin Dhama, Nitin Garg
2. A Primer of Anesthesia- Rajeshwari Subramaniam
3. Drugs in anaesthesia & intensive care- Edward Scarth & Susan Smith

FIRST SEMESTER

APPLIED SURGICAL TECHNOLOGY

Theory: 45 hours

OBJECTIVES:

At the end of the second semester student is expected to understand

1. Common procedures performed in anaesthesia, intensive care unit, and emergency department.
2. Physics and technology involved in the functioning of special equipment used to aid the procedures.
3. Safely transportation of the patient

COURSE OUTCOMES

At the end of the course, students will be able to...

CO1: Understand developing a surgical diagnosis

CO2: Demonstrate and apply aseptic technique.

CO3: Understand and demonstrate suturing techniques CO4:

Classify and identify surgical instruments

CO5: Maintenance and troubleshoot specialized surgical devices

Unit-1

- Developing a surgical diagnosis
    - Complete and thorough history
    - Physical examination
        - Radiographic findings
        - Laboratory investigations
        - Record keeping
        - Differential diagnosis

Unit-2

- Basic necessities for surgery

    - Aseptic technique
        - Medical asepsis
        - Surgical asepsis
    - Pre surgical preparation of the patient
    - Surgical Infection
        - Factors for wound infection
        - Management of abscess
        - Antibiotic prophylaxis
- Infection prevention and universal precautions

- Surgical Incisions
- Hemostasis
    - Means of promoting wound Hemostasis
    - Dead space management

Unit-3

- Decontamination and debridement
- Edema control
- Patient general health and nutrition
- Wound management
    - Prevention of wound infections
- Suture techniques
- Anastomoses
- Abdominal drainage
- Basic surgical and perioperative considerations
    - Antibiotic prophylaxis
    - Prevention of infection in postoperative period

Unit-4

- Patient Safety
    - Admission Procedure
    - Procedure for Safely Transferring the Patient to the Operating Table
    - Positioning
- Surgical Instrumentation
    - Functioning
    - Uses
    - Maintenance
    - Risks and precautions
    - Electrocautery
    - Harmonica
    - Lasers
    - Surgical bio-microscopes
    - X-ray
    - Endoscopes

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Applied surgical technology** shall be as given under.

Type of questions	**No. of questions**	**Marks**	**Total**
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20marks

There shall be no Practical examination

Reference books:

1. Basic surgical skills and techniques- Sudhir Kumar Jain, David L Stoker & Raman Tanwar
2. Berry & Kohn‟s Operating Room Technique (Latest edition)
1. Pocket guide to the Operating Room- Jaypee Latest edition

SECOND SEMESTER

ADVANCED ANAESTHESIA TECHNIQUES
Theory: 45 hours
Practical/Demonstration: 30 hours
OBJECTIVES:

At the end of the second semester student is expected to understand

1. Common procedures performed in anaesthesia, intensive care unit, and emergency department.
2. Physics and technology involved in the functioning of special equipment used to aid the procedures.
3. Understand and manage anaesthetic emergencies in the intraoperative period

COURSE OUTCOMES

At the end of the course, students will be able to...

- CO1: Understand and practice peripheral venous cannulation
- CO2: Demonstrate and prepare vascular cannulation
- CO3: Explain and demonstrate the applications of ultrasound in ICU
- CO4: Understand modern technology in anaesthesia
- CO5: Understand and apply emergency anaesthesia guidelines

Unit- I

- Vascular cannulation
- Central neuraxial blockade
    - Potential benefits of central neuraxial block
    - Mechanism of action, spread, uptake & elimination
    - Ultrasound for central neuraxial blockade
- Peripheral nerve blocks
- Post anaesthesia care
- Ultrasound in ICU
    - FAST
    - Volume assessment
    - Thoracic ultrasound

Unit- II

- Review of modern technology in anaesthesia
    - Ultrasound
    - Fiberoptic
    - X-ray
- Smart Pumps and Computer-Controlled Drug Infusion Delivery
- Anaesthesia information management systems
- Clinical information systems in critical care
- Decision support system

Unit- III

- General anaesthesia
    - Types and techniques
    - Awareness during anaesthesia
    - Complications
    - The long term effects of general anaesthesia
    - Management of general anaesthesia
    - Anesthesia and children

Unit- IV

- Emergency anaesthesia guidelines
    - Incidence and risk factor
- Anaesthetic emergencies
    - Airway emergencies
    - Anaphylaxis
    - Local anaesthetic toxicity
    - Malignant hyperthermia
- The principles and conduct of anaesthesia for emergency surgery
    - Choice of anaesthetic technique
    - Management and protection of the airway including pulmonary aspiration
    - The rapid sequence induction: evolution over time
    - Management of ventilation
    - Maintenance of anaesthesia

Practical/Demonstration

- Assembly of cart for and knowledge of technique of
    - Vascular cannulation
    - Central neuraxial blockade
    - General Anaesthesia
- Assembly of cart for Peripheral nerve blocks
- Use of special equipment and trouble shoot
    - Ultrasound used in anaesthesia
    - Fiberoptic
    - Smart pumps and computer controlled drug infusions
- Identification and management of airway emergencies
- Preparation of anaesthesia cart for emergency surgeries
- Intraoperative maintenance of patient in terms of monitoring and ventilation.

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Advanced Anaesthesia Techniques** shall be as given under.

Type of questions	**No. of questions**	**Marks**	**Total**
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20marks
Practical examination-100marks

Marks pattern for practical examination:-

Practical Exercises	**Viva- Voce**	**Internal assessment**
60	20	20
Grand Total- 100		

Reference books:

1. Step by step practical aspects of emergency anesthesia- Arun Kumar Paul
2. Anesthesiology updates for postgraduates- Sampa Dutta Gupta
3. Morgan and Mikhail's Clinical Anesthesiology

SECOND SEMESTER

SPECIALITY ANAESTHESIA-I

Theory: 45 hours

Practical/Demonstration: 30 hours

OBJECTIVES:

Upon completion of this semester, students will achieve knowledge and level of expertise & proficiency in:

1. Anaesthetic and surgical requirement for different subspeciality procedures in terms of equipment and monitoring.
2. Functioning and procedures of pain clinic.

COURSE OUTCOMES

At the end of the course, students will be able to...

- CO1: Understand, prepare and demonstrate the anaesthesia for Laparoscopic and Robotic surgeries
- CO2: Understand, prepare and demonstrate the anaesthesia for Obstetrics and Gynecological procedures
- CO3: Understand, prepare and demonstrate the anesthesia for Orthopedic surgeries. CO4: Understand, prepare and demonstrate the anesthesia for Ophthalmic surgeries.
- CO5: Understand, prepare and demonstrate the anesthesia for Otorhinolaryngoscopic surgeries
- CO6: Understand, prepare and demonstrate the anesthesia for Onco surgeries

Unit- I

- Anaesthesia for Laparoscopic and Robotic Surgeries
  - Patient preparation
  - Anaesthetic Management
  - Special Considerations
  - Post- operative care
- Anaesthesia for Obstetrics & Gynecological procedures
  - Patient preparation
  - Anaesthetic Management
  - Special Considerations
  - Post-operative care

Unit- II

- Anaesthesia for Orthopedic surgeries
  - Patient preparation, Anaesthetic Management, Positioning, Post-operative care for
  - Upper limb procedures
  - Hip and knee
  - Foot & ankle
  - Pediatric orthopedics
  - orthopedic emergency/trauma
- Anaesthesia for Ophthalmic surgeries

- Patient preparation
- Anaesthetic Management
- Special Considerations
- Post-operative care

Unit- III

- Anesthesia for Otorhinolaryngologic Surgeries
    - Preoperative considerations
    - Intraoperative Management
    - Special Considerations
    - Post-operative care

- Anesthesia for Onco surgeries
    - Preoperative considerations
    - Intraoperative Management
    - Special Considerations
    - Post-operative care

Practical/Demonstration:

- Setting up of monitor, equipment and preparation of the patient for
    - Anaesthesia for Laparoscopic Surgery
    - Anaesthesia for Robotic Surgery
    - Anaesthesia for obstetrics and gynaecological procedures
    - Anaesthesia for ophthalmic procedures
    - Anaesthesia for orthopaedic surgeries
    - Anesthesia for Otorhinolaryngologic Surgeries
    - Anesthesia for Onco surgeries

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Specialty Anesthesia-I** shall be as given under.

Type of questions	**No. of questions**	**Marks**	**Total**
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20marks

Practical examination-100marks

Marks pattern for practical examination:-

Practical Exercises	**Viva- Voce**	**Internal assessment**
60	20	20
Grand Total- 100		

Reference books:

1. Anaesthesiology updates for postgraduates- Sampa Dutta Gupta
2. Textbook of Anaesthesia- Alan R Aitkenhead, David J Rowbotham, Graham Smith
3. A basic textbook is essential - Fundamentals of Anaesthesia- Tim Smith, Colin Pinnock, Ted Lin, and Robert Jones
4. Morgan and Mikhail's Clinical Anesthesiology

SECOND SEMESTER

CLINICAL SURGERY-I
Theory: 45 hours
Practical/Demonstration: 30 hours
OBJECTIVES:

At the end of this semester students should understand

1. Various sub speciality surgical procedures
2. Different requirement for each of these procedures
3. Preparation of patient, equipment, operation theatre for these surgical procedures

COURSE OUTCOMES

At the end of the course, students will be able to...

- CO1: Understand & prepare for Laparoscopic & Robotic surgeries
- CO2: Understand spinal cord injuries and management of spinal trauma
- CO3: Describe types of fractures and its management
- CO4: Understand & prepare for ophthalmic surgeries
- CO5: Understand & prepare for ENT surgeries
- CO6: Understand & prepare for Obstetrics & gynecological procedures

Unit- I
Laparoscopic Surgery

- Principles of laparoscopic surgery
- Advantages and disadvantages
- Safety issues and indications
- Postoperative care
- Laparoscopic Cholecystectomy
- Laparoscopic Appendicectomy
- Advanced Laparoscopic Surgeries
- Diagnostic Laparoscopy

➢ **Robotic surgery**

- Principles of robotic surgery
- Advantages and disadvantages
- Safety issues and indications
- Postoperative care

Unit- II

➢ **Neck and spine**

- The accurate assessment of spinal cord injuries
- The basic management of spinal trauma and major pitfalls

➢ **Trauma to the face and mouth**

- Classification of facial fractures
- Diagnosis and management of fractures

➢ **Elective orthopedics**

Pathology, assessment and management of

- Upper limb
- Hip and knee
- Foot and ankle
- Pediatric orthopedics

Unit- III

- **Ophthalmic surgeries**
  - Cataract extraction
  - Corneal transplant/ keratoplasty
  - Correction of strabismus
- **ENT surgeries**
  - Tympanoplasty
  - Mastoidectomy
  - FESS
  - Rhinoplasty
  - Tonsillectomy & Adenoidectomy
  - Tracheostomy
  - Laryngectomy
  - Radical Neck Dissection
- **Obstetrics & Gynaecological Procedures**
  - D&C
  - Hysterectomy
  - Salphingooophorectomy
  - Tubal Sterilization
  - LSCS
  - Ectopic pregnancy

Practical/Demonstration:

- Setting up of Equipments and preparation of patient for
  - Laparoscopic surgery
  - Robotic surgery
  - Neck and spine surgery
  - Trauma to the face and mouth
  - Elective orthopedic surgery
  - Ophthalmic surgery
  - ENT surgeries
  - Obstetrics and gynaecological procedures

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Clinical Surgery-I** shall be as given under.

Type of questions	No. of questions	Marks	Total
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20marks
Practical examination-100marks

Marks pattern for practical examination:-

Practical Exercises	Viva- Voce	Internal assessment
60	20	20
Grand Total- 100		

Reference books:

1. Bailey & Loves Short Textbook of Surgery
2. Text book of surgery – S. Das
3. Manipal manual of surgery-K. Rajgopal Shenoy & Anitha Shenoy
4. Short text book of surgery – Himasu Roy

SECOND SEMESTER

RESEARCH METHODOLOGY & BIOSTATISTICS

Theory: 60 hours

OBJECTIVES:

At the end of the semester students should be able to

1. Understand and apply statistical methods for the design of biomedical research and analysis of biomedical research data;
2. Learn to participate in a research team in study design, data coordination and management, and statistical analysis and reporting of study results

COURSE OUTCOMES

At the end of the course, students will be able to...

- CO1: Understand & apply scales of measurements, incidence, prevalence and central tendency
- CO2: Understand sampling methods
- CO3: Describe skewness and kurtosis
- CO4: Explain tests of significance
- CO5: Understand correlation & regression
- CO6: Understand & estimate sample size determination CO7: Describe study designs & scientific documentation

Unit 1:

Introduction

Introduction to biostatistics &research methodology, types of variables & scales of measurements, measure of central tendency & dispersion, rate, ratio, proportion, incidence & prevalence

Unit 2:

Sampling

Random and non-random sampling, Different sampling techniques – simple random, stratified, systematic, cluster& multi-stage. Sampling and non-sampling errors and methods of minimizing these errors

Unit 3:

Sampling distribution. Statistics and parameter. Standard error. Basic probability distributions- Normal, Poisson and Binomial distributions with their application in biological sciences. Skewness & Kurtosis.

Unit 4:

Tests of significance

Basics of testing of hypothesis – Null & Alternative hypothesis, type 1 and type II errors, level of significance & power of the tests, p value. Different Parametric Tests – T test (paired & unpaired), & Test for proportion, One-way analysis of variance. Repeated measures analysis of variance. Non-Parametric Tests of significance Chi square test– Mann – Whitney U Test, Wilcoxon Test, Kruskal – Wallis Analysis of variance by ranks, Friedman's test.

Unit 5:

Correlation and regression
Linear correlation by Karl Pearson and Rank order correlation due to Spearman. Testing the significance of correlation. Linear and Multiple regression.

Unit 6:

Sample size determination
General concept. Sample size for estimating means and proportion, testing of difference in means and proportions of two groups.

Unit 7:

Study designs
Descriptive epidemiological methods – case series analysis and prevalence studies. Analytical epidemiological methods – case control and cohort studies. Clinical trials / intervention studies, odds ratio and relative risk, stratified analysis

Unit 8:

Multivariate analysis
Concept of multivariate analysis, introduction to logistic regression and survival analysis

Unit 9:

Reliability and validity of diagnostic tests

Unit 10:

Scientific documentations
Structure of research protocols, structure of thesis/research report, formats of reporting in scientific journals. Systematic review and meta-analysis.

Scheme of Theory Examination:
There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Research methodology and Biostatistics** shall be as given under.

Type of questions	No. of questions	Marks	Total
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20marks
There shall be no practical examination

Reference books:

1. ABC of research methodology and applied biostatistics by MN Parick & Nithya Gogtay.
2. Introduction to biostatistics and research methods by P.S.S. Sundar Rao & J. Richard
3. Research methodology & Biostatistics- A comprehensive guide for health care professionals-Suresh K. Sharma
4. Guide to research methodology and Biostatistics-KMK Masthan

THIRD SEMESTER

SPECIALITY ANAESTHESIA- II

Theory: 45 hours

Practical/Demonstration: 30 hours

OBJECTIVES:

Upon completion of this semester, students will achieve knowledge and level of expertise & proficiency in:

1. Anaesthetic and surgical requirement for different sub speciality procedures in terms of equipment and monitoring.
2. Functioning and procedures of pain clinic.
3. Basic legal ethical issues in organ transplant

COURSE OUTCOMES

At the end of the course, students will be able to...

- CO1: Understand, prepare and demonstrate the anaesthesia for Neurosurgeries
- CO2: Understand, prepare and demonstrate the anaesthesia for Pediatric surgeries
- CO3: Understand, prepare, and demonstrate the anesthesia for Cardiovascular surgeries.
- CO4: Understand, prepare, and demonstrate the anesthesia for Genitourinary surgeries.
- CO5: Understand, prepare, and demonstrate the anaesthesia for Plastic & reconstructive surgeries.
- CO6: Understand, prepare and demonstrate the anaesthesia for organ transplantation

Unit- I

- Anaesthesia for Neurosurgeries
  - ICP & ICT
  - Patient preparation
  - Anaesthetic Management
  - Special Considerations
  - Post- operative care

- Anaesthesia for Paediatric surgeries
  - Patient preparation
  - Anaesthetic Management
  - Special Considerations
  - Post-operative care

Unit- II

- Anaesthesia for Cardiovascular surgeries
    - Patient preparation
    - Anaesthetic Management
    - Positioning
    - Post-operative care
    - Cardiopulmonary bypass

- Anaesthesia for genitourinary surgeries
    - Patient preparation
    - Anaesthetic Management
    - Special Considerations
    - Post-operative care

Unit- III

- Anaesthesia for plastic & reconstructive Surgeries
    - Preoperative considerations
    - Intraoperative Management
    - Special Considerations
    - Post-operative care

- Anaesthesia for organ transplantation
    - What is organ transplantation
    - The transplant process
    - Cadaveric organ donation
    - Living organ donation
    - Anaesthetic management

Practical/Demonstration:

- Setting up of monitor, equipment and preparation of the patient for
    - Anaesthesia for Neurosurgeries
    - Anaesthesia for Paediatric surgeries
    - Anaesthesia for Cardiovascular surgeries
    - Anaesthesia for genitourinary surgeries
    - Anaesthesia for plastic & reconstructive Surgeries
    - Anaesthesia for organ transplantation

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Specialty Anesthesia- II** shall be as given under.

Type of questions	**No. of questions**	**Marks**	**Total**
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20marks

Practical examination-100marks

Marks pattern for practical exams:-

Practical Exercises	**Viva- Voce**	**Internal assessment**
60	20	20
Grand Total- 100		

Reference books:

1. Anaesthesiology updates for postgraduates- Sampa Dutta Gupta
2. Textbook of Anaesthesia- Alan R Aitkenhead, David J Rowbotham, Graham Smith
3. A basic textbook is essential - Fundamentals of Anaesthesia- Tim Smith, Colin Pinnock,Ted Lin, and Robert Jones
4. Morgan and Mikhail's Clinical Anesthesiology

THIRD SEMESTER

CLINICAL SURGERY-II
Theory: 45 hours
Practical/Demonstration: 30 hours

OBJECTIVES:

At the end of this semester students should understand

- Various sub speciality surgical procedures
- Different requirement for each of these procedures
- Preparation of patient, equipment, operation theatre for these surgical procedures

COURSE OUTCOMES

At the end of the course, students will be able to...

- CO1: Understand & prepare for Cardiac, thorax and vascular surgeries
- CO2: Understand organ transplantation procedures
- CO3: Describe Elective neurosurgeries
- CO4: Understand & prepare for plastic & reconstructive surgeries

Unit- I

- Elective neurosurgery
  - Head injury
  - Investigation and treatment for intracranial infection
  - Treatment for hydrocephalus
  - Management of intracranial hemorrhage
  - Management of epilepsy
  - Understanding the principles involved in brain death

- Plastic and reconstructive surgery
  - The spectrum of plastic surgical techniques
  - The various skin grafts
  - The principles and use of flaps
  - Plastic surgery to manage difficult and complex tissue loss

Unit- II

- The thorax
  - Investigation of chest pathology
  - Surgical oncology as applied to chest surgery

- Vascular surgery
  - Investigation for vascular surgery
  - Management technique of vascular surgery
    - Direct repair by stunting
    - Endarterectomy
    - Bypass

- Cardiac surgery
    - The role of investigation in planning of surgery
    - The management of coronary heart disease
    - The role of surgery in valvular heart disease
    - Special role of surgery in congenital heart disease
    - The management of aortic vascular and pericardial disease

Unit- III

- Organ Transplantation
    - What is organ transplantation
    - The transplant process
    - Timeline of medical and legal advances in organ transplantation
    - Cadaveric organ donation
    - Living organ donation
    - Alternative organs
    - The impact of transplantation

Practical/Demonstration:

- Setting up of Equipments and preparation of patient for
    - Elective neurosurgery
    - Plastic and reconstructive surgery
    - The thoracic surgery
    - Vascular surgery
    - Cardiac surgery
    - Organ Transplantation

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Clinical Surgery-II** shall be as given under.

Type of questions	**No. of questions**	**Marks**	**Total**
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20marks

Practical examination-100 marks

Marks pattern for practical examination: -

Practical Exercises	**Viva- Voce**	**Internal assessment**
60	20	20
Grand Total- 100		

Reference books:

1. Bailey & Loves Short Textbook of Surgery
2. Text book of surgery – S. Das
3. Manipal manual of surgery-K. Rajgopal Shenoy & Anitha Shenoy
4. Short text book of surgery – Himasu Roy

THIRD SEMESTER

HEALTH CARE MANAGEMENT
Theory: 45 hours
OBJECTIVES:
At the end of the third semester students should be able to

1. Discuss the role of the manager in healthcare and how organizations and people work within the healthcare system.
2. Effectively manage people, finances and organizational resources.
3. Complete an organizational development project, reflect on the learning gained and evaluate the project from a leadership and healthcare management perspective.

COURSE OUTCOMES

At the end of the course, students will be able to...

- CO1: Understand & apply management concepts & theories
- CO2: Mention management functions & process
- CO3: Describe basics of HRM & sourcing
- CO4: Understand staff training & development
- CO5: Understand materials management
- CO6: Describe Inventory control, value analysis & biomedical waste management

Unit-I

- **Management concepts and theories**
  - Management and organizations
  - Management role
  - Levels of managers and management skills
  - Classical school
  - Behavior school
  - Management science school

- **Management functions and process**
  - Planning
  - Organizing
  - Staffing
  - Directing
  - Controlling

Unit- II

- **Basics of HRM and sourcing**
  - Introduction and relationship between HRM and HRD
  - Objectives of HRM
  - HR planning: short term and long term
  - Productivity analysis in healthcare
  - HR policy and procedure
  - Recruitment
  - Selection
  - Placement
  - Induction / Orientation
- **Training and development**
  - Staff training and development
  - Career growth and development
  - Management development

Unit- III

- **Materials management**
  - Introduction
  - Definition and function
  - Goals and objectives of materials management
  - Problems and issues in hospitals
- **Equipment purchase and maintenance**
  - Planning and selection of equipment
  - Import of equipment
  - Equipment utilization and operation
  - Equipment repair and maintenance
  - Equipment audit

Unit- IV

- **Scientific inventory management**
  - Codification and standardization
  - Value analysis
  - Inventory control
  - Lead time, safety stock and reorder level
  - Economic order quantity (EOQ)
  - Selective controls
  - Case studies on inventory control
  - The biomedical waste (management and handling) rules

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Health care management** shall be as given under.

Type of questions	**No. of questions**	**Marks**	**Total**
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20marks

There shall be no practical examination

Reference books:

1. Introduction to health care management by Sharoon B & Nancy H
2. Foundations of health care management – Bernard J Healy & Marc C Marchese
3. Dunn & Haimann‟s Health Care Management

FOURTH SEMESTER

CLINICAL MEDICINE & CRITICAL CARE MANAGEMENT
Theory: 75 hours
Practical/Demonstration: 30 hours

OBJECTIVES:

- At the end of the fourth semester students should be able to
- Overall idea of ICU care, emergency care, and transport of patients.
- Knowledge of techniques, procedures and monitoring of ICU patients.
- General care, nutrition and respiratory care of ICU patients.
- Basic Life Support and some knowledge of ACLS.
- Understand the emergency/ disaster management cycle
- Develop a basic knowledge of prevention, mitigation, preparedness, response and recovery in disaster
- Have a basic understanding emergency management.
- Resuscitation and triage skills

COURSE OUTCOMES

At the end of the course students should be able to…

- **CO1:** Understand universal precautions
- **CO2:** Report patient vital signs with different monitoring systems
- **CO3:** Know the techniques, procedures and treatment of intensive care unit patients
- **CO4:** Understand general care, nutrition and respiratory care of intensive care unit patients
- **CO5:** Learn to manage the hospital acquired infections
- **CO6:** Participate in disaster and triage management and trauma management
- **CO7:** Perform basic life support (BLS) and advanced cardiac life support (ACLS)
- **CO8:** Learn to monitor the patient during inter hospital and intra hospital transport

Unit-I

- First aid for unconsciousness
    - Aims, principles & rules of first aid
    - First aid box
- Advance Trauma Life Support
- Cardio pulmonary resuscitation
- Basic life support
    - Algorithm
    - Mouth to mouth ventilation
    - External cardiac compression
- ACLS

- Defibrillation
- Vascular access
- Definitive airway
- Foreign body obstruction
- Drugs

- CPR in infants and children
- Complications of BLS

Unit- II

- Monitoring techniques in ICU practice
  - Invasive blood pressure (BP) monitoring
  - Transoesophageal Doppler (TED)
  - Measurement of central venous pressure (CVP)
  - Pulmonary artery catheterization
  - Arterial blood gas (ABG) analysis
- Ventilator Life Support in ICU
  - Working principles of ventilator in ICU
  - Types of ventilators
  - Mechanical ventilation modes and settings
  - Ventilator management
  - Ventilation induced lung injury
  - Ventilation monitoring
  - Non-conventional ventilation
  - Weaning from the ventilator
  - Spontaneous Awakening Trial (SAT) & Spontaneous Breathing Trial (SBT)

Unit- III

- Intubation and tracheostomy
- Spirometry - Data analysis
- Fluid control and therapy
- Supportive care
  - Control of infection
  - Transport of critically ill
  - Investigations
- Patient Safety Bundle
  - Ventilator Care Bundle
  - Central-line Bundle
  - Severe Sepsis Bundle

Unit- IV

- Hospital disaster preparedness and response
  - Coordination and management
  - Planning, training

- Information, communication and documentation
- Medico legal concerns
- Safety and security
- Human resources
- Triage
- Post disaster recovery
- Patient handling

Unit- V

- Neurologic evaluation or monitoring
  - Sedation score
  - Glasgow coma scale
- Shock and Multisystem failure
  - Sepsis
  - Septic shock
  - Hemorrhagic shock
  - Neurogenic shock
  - Anaphylactic shock
  - Acute Lung Injury (ALI) and Acute Respiratory Distress syndrome (ARDS)
  - Multi-organ dysfunction syndrome

Practical/Demonstration: **30 hours**

- Principles and setting up of
  - Invasive blood pressure monitoring 02 hours
  - Transesophageal Doppler 01 hours
  - Measurement of Central Venous pressure 02 hours
  - Pulmonary artery catheterization 01 hours
  - Arterial blood gas analysis 01 hours
  - Intracranial pressure measurement 01 hours
  - Intra-abdominal pressure measurement 01 hours
- Working principle and setting up of ventilator 03 hours
- Hand washing and universal precautions 02 hours
- Transport of critically ill patient 02 hours
- Training on- First aid, BLS, ACLS, ATLS and defibrillation 10 hours
- Mock drill on disaster preparation 04 hours

Scheme of Theory Examination:

There shall be one theory paper of **three** hours duration carrying **80** marks. Distribution of type of questions and marks for **Clinical Medicine & Critical care management** shall be as given under.

Type of questions	**No. of questions**	**Marks**	**Total**
Long Essay(LE)	02	15	30
Short Essay(SE)	10	05	50
Grand Total			**80**

Theory Internal Assessment-20marks

Practical examination-100marks

Marks pattern for practical examination: -

Practical Exercises	**Viva- Voce**	**Internal assessment**
60	20	20
Grand Total- 100		

Reference books:

1. Civetta, Taylor & Kirbey‟s Critical care
2. Critical care- Paul Marino
3. The ICU book-Schumacher

FOURTH SEMESTER

DISSERTATION /PROJECT WORK

COURSE OUTCOMES:

At the end of the course students will be able to

- **CO1:** Develop the ability to apply the methods while working on a research project work
- **CO2:** Describe the appropriate statistical methods required for a particular research design
- **CO3:** Choose the appropriate research design and develop appropriate research hypothesis for a research project
- **CO4:** Develop an appropriate framework for research studies

The dissertation work is aimed at training a postgraduate candidate in research methodology and techniques. It includes identification of the problem, formulation of a hypothesis, review of literature, getting acquainted with recent advances, designing of a research study, collection of data, critical analysis, and comparison of results and drawing conclusions.

Preparation of dissertation

The written text of dissertation shall be prepared as per the respective university guidelines.

Scheme of evaluation

Dissertation shall be evaluated at the time of university examination of 4thsemester by the panel of examiners (internal & external) appointed by the university. He should obtain a score of 50% aggregate to be declared pass in IV semester examination.

Evaluation format for dissertation

Sl. No	University Evaluation	Marks			
		Max Marks	Internal Examiner	External Examiner	Marks obtained
1.	Objectives, Research Question, Literature Review	25			
2.	Results and Discussion	25			
3.	Dissertation Defence	25			
4.	Viva voce by Internal and External Examiner	25			
	Total	**100**			

1.12. Internship Logbook Performa

THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF MEDICAL & ALLIED SCIENCES

Department of Anesthesia & Operation Theatre Technology

STUDENT'S LOG BOOK
For
Masters in Anesthesia & Operation Theatre Technology



Student Name:

……

Year & Month.:

……

Registration No.:

……

Guidelines for Logbook Entry Outcome:

Outcome - Morbidly / Mortality / Any major complication should be indicated with measures taken for timely management.

i. The candidate should make the entries in the log book daily, countersigned by the Assistant Professor / Associate Professor / Professor / Head of the Department.

ii. The Associate Professor / Professor must review & sign the log book every week for short-duration postings & every month for long-duration postings

iii. In emergency OT posting the log book will be countersigned by Senior Resident on duty.

iv. Any classes taken for BAOTT students or time spent on a thesis should be explicitly noted along with the nature of work in the log book.

v. The logbook is to be submitted through the Professor / Associate Professor at the time of the final examination.

vi. Please use clear handwriting to fill the Logbook.

Area of Posting…… Period……to……

Date						
Patient ID						
Age/Sex						
Diagnosis						
Operative Procedure						
Anesthetic Technique						
Special Procedure						
In time						
Out time						
Observation/Work done/Newly learned things						
Any Classes Taken for BAOTT students						
Time, Nature of work regarding thesis work.						
Signature of Faculty/Senior Resident						

1.13. Competencies Statement:

Master in Anaesthesia & Operation Theatre Technology

Learning Objectives:

At the completion of this course, the student should be able to –

1. Provide advanced nursing care to manage the complex clinical needs of patients in an intensive care unit and operating room
2. Managing patient care procedures during anaesthesia
3. Various anaesthesia techniques- theory and skills, anaesthetic drugs, specialty anaesthesia with complications and management under supervision of the Anaesthesiologist.
4. Understanding monitoring needs, preventing, detecting complications using technologically advanced monitors & instruments and treating the condition respecting the limitation of his own competence
5. Capable to order and interpret full spectrum of diagnostic tests and special procedures related to care of critically ill, under the supervision of a critical care specialist and evaluate the outcomes of intervention.
6. The application of relevant theories, research, and evidence-based guidelines, and meet psychological needs of patients in critical care situation
7. Managerial skills, effective communication with patients ,relatives, medical staff in high stress conditions, working within norms and regulation as well as ethics and following deontological principles.

Expectation from the future post graduates in the providing patient care

1. Student will be able to demonstrate comprehensive knowledge covering the main areas of Anaesthetic management of the all critical care medicine.
2. Student will be able to apply clinical reasoning and integrate knowledge with practice in a wide, and often unpredictable, variety of clinical conditions.
3. You will be able to demonstrate criticality and problem-solving in the areas of academic and clinical critical care.
4. Student will be able to manage patient care procedures during anaesthesia and critical care
5. Student will have ability to work independently and take responsibility for his own work, collaborating in activities of clinical research and training.


System based practice
Informatics and Technology
Quality Care and Patient safty
Person-Centred Care
Communication and Teamwork
Evidence based parctice
Professionlism and leadership

1.14. Skills based outcomes and monitorable indicators:

Performance Criteria	Indicators		
	Knowledge	**Skill**	**Behavior**
• **Quality care and patient safety**	• Should have in depth knowledge about various general health conditions while examination • Should have essential knowledge about providing care to each individual who visits to your hospital • Should be aware of different types of disabilities and information on patients with additional need. • Should have essential knowledge about rights and dignity of the patients.	• Proficiency in assisting surgeons/anaesthetist with equipment‟s, machines instruments, and suturing. • Competence in aiding anesthetists with anesthesia induction, airway management, and vital sign monitoring.	• Proactive attitude in preparing for procedures and anticipating the needs of the surgical team/Anesthesia team. • Attention to detail, ensuring proper organization of instruments and supplies. • Effective time management and collaboration with the surgical team. • Adaptability to handle unexpected situations and adjustments during surgery. • Professionalism, including confidentiality, ethical conduct, and respect for boundaries.
• **Quality care and patient safety**	• Thorough understanding of the principles of aseptic technique and sterile field maintenance.	• Proficiency in performing hand hygiene using appropriate handwashing or hand sanitization methods.	• Maintaining appropriate hand hygiene procedures both inside and outside the operating room.

	• Knowledge of infection control guidelines, including hand hygiene, proper gowning and gloving techniques, and disinfection protocols. • Familiarity with different types of sterilization methods, such as steam sterilization, ethylene oxide sterilization, and sterile packaging	• Competence in donning sterile gowns and gloves without contaminating them and in maintaining sterility while working within the sterile field. • Skill in handling sterile instruments, supplies, and drapes to prevent contamination and maintain a sterile environment. • Ability to assist surgeons and anesthetists in maintaining sterility during procedures, such as passing instruments or opening sterile packages.	• Diligently adhering to the recommended gowning and gloving procedures to protect oneself and the patient. • Pay close attention to protecting the sterile field's integrity by limiting air circulation, preventing needless movements, and contaminating it with non-sterile things. • Strict adherence to sterile procedure during instrument handling, making sure that instruments stay in the sterile field and maintaining the proper sterile instrument passes. • Addressing any breaches of sterile technique or potential contamination as soon as possible, taking corrective action, and, as necessary, informing the surgical team. • Maintaining sterility during the procedure with constant attention and awareness while aggressively looking out for any potential breaks or compromises.

• **System based practice**	• Understanding of surgical instruments and their uses. • Familiarity with medical equipment, such as anesthesia machines and monitors and Troubleshooting • Knowledge of equipment setup and placement.	• Proficiency in handling surgical instruments and ensuring their sterility. • Ability to set up and place medical equipment accurately. • Skill in operating equipment and adjusting settings. • Troubleshooting abilities for equipment malfunctions.	• Continuously update knowledge of instruments and equipment. • Ensure instruments are clean and organized. • Double-check equipment setup for patient safety. • Communicate effectively with surgeons and anesthetists. • Promptly address equipment issues during procedures. • Collaborate with appropriate personnel for complex troubleshooting.
• System based practice	• Understanding of surgical procedures, including laparoscopic, arthroscopic, and robotic techniques. • Knowledge of different surgical instruments, sutures, and their uses. • Familiarity with sterile techniques and maintaining a sterile field.	• Proficiency in directly supporting the surgeon during procedures. • Ability to anticipate the surgeon's needs and provide timely assistance. • Skill in passing sutures, instruments, and other necessary items accurately.	• Adhere to the surgeon's directions promptly and accurately. • Anticipate the surgeon's demands and prepare instruments or sutures in advance. • Maintain clear and open communication with the surgeon throughout the procedure.

		• Competence in scrubbing for more technical procedures like laparoscopic, arthroscopic, and robotic surgeries.	• Collaborate with the surgical team to ensure effective teamwork. • Demonstrate attentiveness and responsiveness during the surgery. • Continuously enhance knowledge of surgical procedures and instruments. • Adhere to sterile techniques and maintain a sterile environment.
• System based practice (Anesthesia Support)	• Understanding of anesthesia administration and management. • Familiarity with the different stages of anesthesia induction and maintenance. • Awareness of monitoring vital signs, oxygen saturation, and end-tidal carbon dioxide levels taking ABG sample, BIS, NMT, Arterial line/CVP line insertion.	• Proficiency in assisting with the administration and supervision of anaesthesia. • Ability to assist in securing and positioning patients during anesthesia induction. • Ability to monitor and assess vital signs, oxygen saturation, and end-tidal carbon dioxide levels.	• Collaborate closely with the anaesthetist, following their instructions and assisting with anaesthesia administration. • Help secure and position patients during anaesthesia induction and throughout the procedure. • Pay close attention to any changes or irregularities in patient conditions. • Promptly communicate any significant changes to the anaesthetist and surgical team.

• Communication and Teamwork.	• Understanding of emergency procedures and protocols. • Knowledge of the locations and functions of emergency equipment/crash carts. • Familiarity with potential postoperative complications and their management.	• Proficiency in identifying and accessing emergency equipment quickly. • Ability to remain calm and composed in high-stress situations. • Competence in following established emergency protocols. • Skill in assisting the surgical team during crises or postoperative difficulties.	• Proactively learn about emergency procedures and equipment locations. • Follow established protocols and guidelines for emergency situations. • Assist the surgical team promptly and effectively during crises. • Continuously update knowledge of emergency procedures and best practices.
• Informatics and technology	• Should have adequate knowledge of data protection and how this will impact security, access and confidentiality of the patient‟s records • Should have essential knowledge to ensure the patient environment will remain safe and user-friendly, in terms of access and facilities	• Conversant in using various digital devices, access cloud storage platforms and saves electronic medical records on system-based software programs and keeps them safe • Demonstrates how to store and retrieve manual medical records	• Seeks consent of the patient before providing information to external stake holders • Restricts self from discussing patient information and condition in any open forum/external communication

• Professionalism and Leadership	• Should have essential knowledge of how to maintain practice in accordance with other professional health care standards	• Explains the condition that are treatable/correctable beyond your practice standards	• Honesty and understanding of own limitations
• Professionalism and Leadership	• Should have in depth knowledge of ethical practice and standard operating procedures followed in the clinical examination	• Explains the uses of various diagnostic instruments and their importance in the process of examination	Can demonstrate and teach of bachelor courses students
• Evidence Based practice	• Should have vital knowledge of the law, codes and guidelines set by the regulatory body of profession and is fully aware of the consequences if not followed.	• Follows the code of conduct set down by the council/ appropriate authorities	Can demonstrate and teach of bachelor courses students